

DiscountMate – GA4 Event Audit Report

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Workstream: Behavioural Analytics & Persona Segmentation (GA4)

Task: DA-01-T1 Audit GA4 Events

1. Introduction

The purpose of this audit was to review the existing Google Analytics 4 (GA4) implementation for the DiscountMate web application. The audit focuses on identifying the behavioural events currently being tracked, evaluating whether these events provide useful information for behavioural analytics and customer persona segmentation, and identifying any gaps where additional tracking could improve future machine learning models.

The findings from this audit will support the Behavioural Feature Specification (DA-01-T3) and provide input for the Behavioural Data Dictionary (DA-01-T7).

2. Audit Scope

Reviewed reports:

- Reports
- Events Report
- Pages & Screens Report
- Realtime Dashboard

The audit was performed using the existing DiscountMate GA4 property over the default reporting period of the last 28 days

3. Current Tracked Events

Event	Count	Purpose	Useful?
page_view	293	Tracks page visits	Yes
scroll	131	Measures scrolling	Yes
session_start	106	New session	Yes
first_visit	75	First-time visitor	Yes
user_engagement	67	Measures engagement	Yes
Home_Page	26	Home page viewed	Yes
Specials_Page	10	Specials page viewed	Yes
Dashboard_Page	4	Dashboard viewed	Limited
Compare_Page	2	Comparison page viewed	Yes
MyLists_Page	2	Shopping list viewed	Yes
view_search_results	2	Search results viewed	Yes
Profile_Page	1	Profile viewed	Limited

4. Pages & Screens Analysis

Page	Views	Observation
Home Page (/)	142	Primary landing page with the highest traffic.
Compare (/compare)	27	Highest average engagement time (7m 52s), indicating users spend considerable time comparing products.
Specials (/specials)	24	Frequently visited, showing strong user interest in discounts and promotions.
Search (/search)	16	Indicates active product search behaviour.
Login (/login)	15	Standard authentication page.
Register (/register)	10	Indicates new user registrations.
My Lists (/my-lists)	6	Demonstrates usage of shopping list functionality.
Profile (/profile)	6	User profile management page.

5. Behavioural Insights

Based on the audit, the current GA4 implementation already captures several useful behavioural signals:

- Website navigation through page views.
- User engagement using scroll depth and engagement events.
- New and returning user activity.
- Product comparison behaviour.
- Search activity.
- Interest in promotional offers.
- Shopping list interactions.

The Compare page showed the highest average engagement time, suggesting that users spend significant time evaluating products before making purchasing decisions. This behaviour is likely to be valuable for future customer segmentation.

6. Gap Analysis

Suggested Event	Reason
Product_Click	Identify popular products
Add_To_Favourites	Measure long-term preferences
Price_Alert_Created	Identify price-sensitive users
Retailer_Selected	Understand retailer preference
Coupon_Click	Measure promotion engagement
Filter_Used	Understand search refinement
Sort_Used	Understand shopping priorities
Outbound_Merchant_Click	Measure purchase intent

These recommendations are based on the current event list reviewed during the audit and are intended to improve future behavioural analytics.

7. Recommendations

Based on the findings, the following recommendations are proposed:

1. Continue using the existing GA4 implementation as the baseline for behavioural analytics.
2. Introduce additional custom GA4 events to capture richer shopping behaviour.
3. Align future event tracking with the Behavioural Feature Specification to ensure the collected data supports clustering and persona segmentation.
4. Maintain documentation of all tracked events within the Behavioural Data Dictionary.

8. Conclusion

The DiscountMate GA4 implementation currently provides a strong foundation for analysing user behaviour. Core engagement metrics, page interactions, search activity, and comparison behaviour are already being collected. These events provide valuable information for understanding customer behaviour and support the initial stages of behavioural analytics.

However, capturing additional shopping-specific events such as retailer selection, product interaction, and price alert creation would provide richer behavioural signals and improve the quality of future customer persona segmentation and clustering models.

Appendix

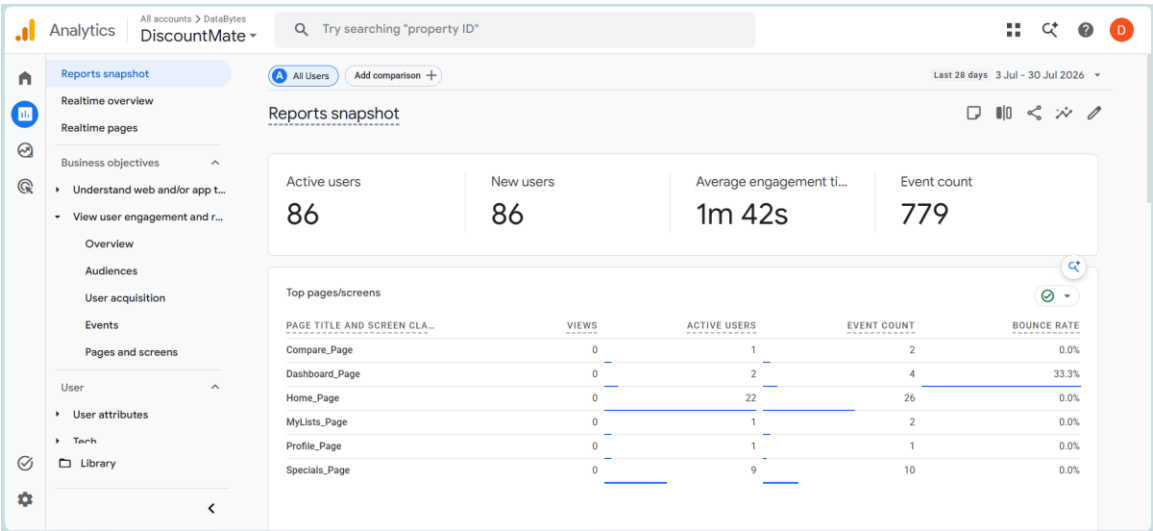


Figure 1. DiscountMate GA4 Reports Snapshot showing user activity, engagement metrics, and overall event summary for the reporting period.

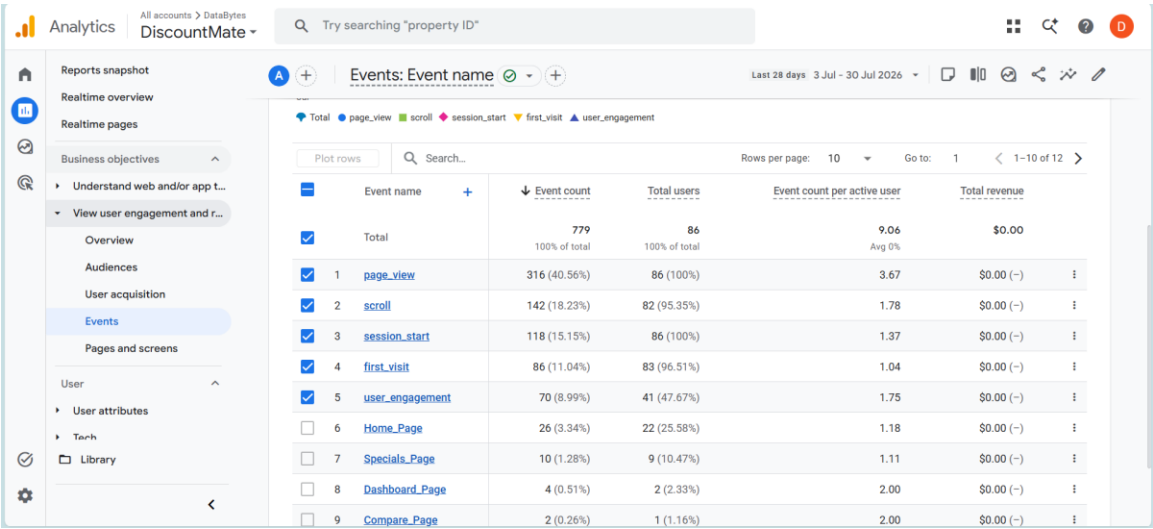


Figure 2. GA4 Events Report displaying the primary behavioural events currently tracked within the DiscountMate application.

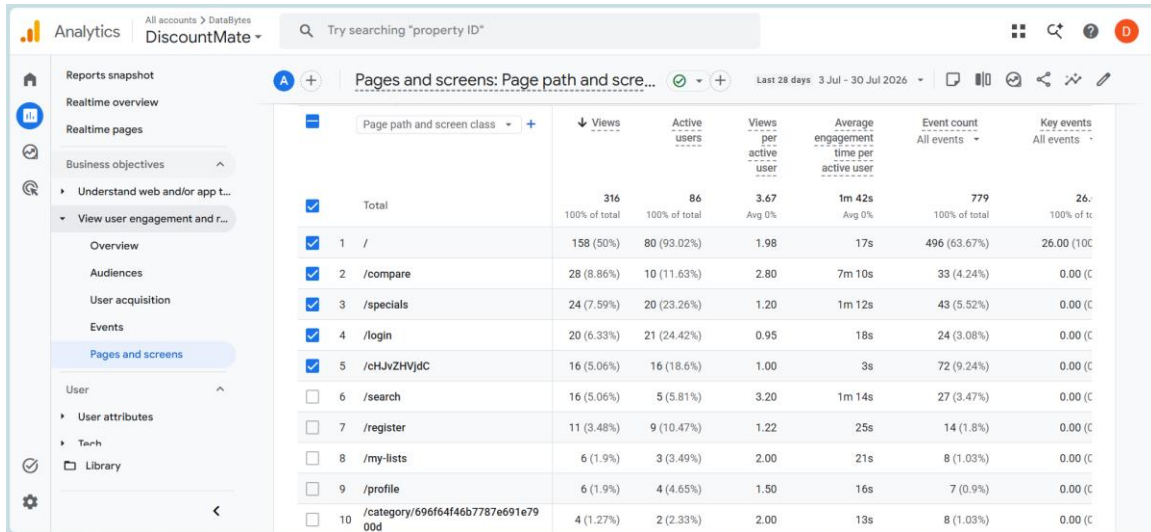


Figure 3. Pages & Screens report highlighting high user engagement on the Compare page, suggesting significant user interaction during product comparison.